

Question:

What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
 $KE = hc / \lambda$, $h = 4.14 \times 10^{-15}$ eV·ns ($h = c \cdot \hbar / 2\pi$), $c = 3 \times 10^{10}$ cm/sec
 $\lambda = hc / KE$
 $= 4.14 \times 10^{-15} \text{ eV} \cdot \text{ns} \cdot 3 \times 10^{10} \text{ cm/sec} / 1 \text{ eV}$
 $= 1.242 \times 10^{-4} \text{ cm} \approx 1.23 \times 10^{-7} \text{ cm}$
</think><answer>G. 1.23×10^{-7} cm</answer>

